

ZERO-COST HACKATHON INFRASTRUCTURE BLUEPRINT

Implementation Roadmap: 100% Free Open-Source AQI & HCHO Tracking

1. INFRASTRUCTURE STRATEGY: TOTAL FINANCIAL FREEDOM

This updated framework ensures that your team can deliver an identical level of technical depth, cloud computation speed, and algorithmic complexity without incurring any costs. By substituting proprietary toolsets with active modern open-source modules and running models on tier-one public cloud accelerator pipelines, the entire platform operates inside a zero-budget profile perfectly scaled for competitive hackathon deployment parameters.

ASSET LAYER	PAID / COMMERCIAL OPTION	100% FREE OPEN-SOURCE ALTERNATIVE
Compute Engines	Local Heavy GPU Rig / AWS EC2 Instances	Google Colab Free Tier / Kaggle Notebooks (Free T4 Tensors)
Statistical Engine	MATLAB Spatial Analytics Toolboxes	SciPy + PySal + Statsmodels (Native Python Libraries)
Geospatial Mapping	ArcGIS Server / Enterprise Engines	QGIS Desktop Application + Google Earth Engine Cloud API
Dashboard Hosting	AWS Elastic Beanstalk / Paid Hosting Tiers	Streamlit Community Cloud / GEE Web Apps

2. STEP-BY-STEP EXECUTION SEQUENCE

Step 1: Environment Setup & Public Cloud Integration

- **Sign Up for Google Earth Engine (GEE):** Access petabytes of geospatial storage for free through a standard non-commercial or academic hackathon developer registration.
- **Initialize Cloud Notebooks:** Create clean project spaces inside **Google Colab** or **Kaggle Kernels**. This provides immediate, out-of-the-box free access to shared NVIDIA T4 GPU graphics cards to speed up multi-dimensional neural matrix configurations.
- **Install Python Modules:** Spin up the runtime cell and run the following command to initialize all required open-source data manipulation frameworks:

```
pip install earthengine-api xarray rasterio geopandas torch pytorch-lightning pysal scikit-learn streamlit
```

Step 2: Data Ingestion & Spatio-Temporal Interfacing

- **Pull Public Satellite Data Streams:** Leverage free cloud API connections to extract Sentinel-5P TROPOMI column values ($HCHO$, NO_2 , SO_2 , CO , O_3) directly from GEE. Establish free developer data links to ISRO's MOSDAC portal for INSAT-3D Aerosol Optical Depth (AOD) configurations and NASA FIRMS for thermal telemetry counts.
- **Acquire Open-Access Ground Truths & Meteorology:** Query the Central Pollution Control Board (CPCB) CAAQM public endpoints to pull point validation records. Concurrently map free atmospheric data layers (wind tracks, surface layers, boundary profiles) out of Copernicus ERA5 datasets.
- **Collocate and Patch Data:** Use free utilities like ``xarray`` and ``geopandas`` to resample geographic boundaries into standardized layout projections. Align shapes over physical station arrays, pulling localized 5×5 pixel bounding grids around coordinates to safeguard sequential deep learning modules from wind dispersion and horizontal mapping inaccuracies.

ZERO-COST INGESTION & INTERFACING SEQUENCE

Query Free GEE API & NASA
FIRMS Feeds



Fetch Open CPCB Station
Truths + ERA5 Met Data



Apply Xarray Resampling to
5x5 Matrix Patches

Step 3: Core Model Development (Objective-1 Pipeline)

- **Structure Input Arrays:** Arrange data arrays inside Google Colab's local volatile storage as 5-dimensional tensors shaped as: $[Batch, Time_Steps, Channels, Height, Width]$.
- **Build the Hybrid Machine Learning Network:** Implement your layers via the open-source PyTorch framework. Route arrays into standard ``TimeDistributed(Conv2D)`` layers to parse incoming visual arrays, and tie outputs straight into ``Bidirectional LSTM`` modules to isolate multi-day temporal accumulation trends.
- **Train with Custom Loss Functions:** Optimize training performance entirely inside your free cloud environment by setting up your compound loss formula: $Loss = (\alpha \times MSE) + ((1 - \alpha) \times MAE)$. Track generalization thresholds across spatial dimensions via ``GroupKFold`` cross-validation grouped cleanly by CPCB Station IDs.
- **Calculate Surface AQI:** Write a standard python routine to parse estimated near-surface arrays against standard CPCB breakpoint parameters, declaring the final absolute cell rating to match the maximum sub-index evaluated across the criteria.

OBJECTIVE-1 MACHINE LEARNING FLOW

5D Tensor Allocation
(Google Colab T4 Memory)



Spatial/Temporal Execution
PyTorch Conv2D + BiLSTM



Index Derivation
Max Sub-Index Check

Step 4: Spatial Analytics & Plume Tracking (Objective-2 Pipeline)

- **Establish Background Baselines:** Isolate true regional biomass-driven tracking signatures by partitioning raw data arrays against moving multi-season baseline profiles.
- **Run Free Spatial Clustering:** Substitute costly commercial analytics software by using the Python **PySAL** library to execute Getis-Ord Gi* hotspot grouping. Extract meaningful regional plume anomalies using strict public filter settings: *Z-score* > 2.58 and *p-value* < 0.01.
- **Execute Trajectory Mapping:** Evaluate and cross-reference U and V atmospheric wind vector channels tracked specifically at the 850 hPa pressure altitude plane to construct downstream plume propagation lines from agricultural point sources across the Indo-Gangetic Plain.

Step 5: Hackathon "Wow Factor" & Dashboard Deployment

- **Compute Free Spatial SHAP:** Load the open-source `shap` library inside your environment to calculate feature importance maps over your spatial layers. This shows the judges exactly how your model weights specific satellite metrics vs. meteorological layers.
- **Deploy on Streamlit Community Cloud:** Push your code repository directly from a public GitHub account into the **Streamlit Community Cloud** platform for free deployment. This provides judges with a public URL where they can interact with timeline sliders, wind trajectories, and downscaled grids without any runtime costs.

Strategic Presentation Note: Highlighting to the evaluating panel that your system runs entirely on scalable, zero-cost public data streams and open-source infrastructure significantly reinforces the viability, scalability, and societal value of your deployment strategy.